NIRVANA 2026 · PeaceOfCode Hackathon

DERMALERT

Early Detection of Neurological Disorders

Track 2 · Team DermAlert

Problem Statement

Track 2 – Early Detection of Neurological Disorders

The Challenge

Neurological disorders such as Alzheimer's Disease (AD) & Mild Cognitive Impairment (MCI) often remain undiagnosed in early stages due to subtle clinical manifestations. Manual MRI interpretation is time-consuming, requires specialized expertise, and is often unavailable in resource-constrained settings.

Task 1	Task 2	Task 3	Task 4
Standardized MRI Preprocessing Pipeline	Binary Classification CN vs AD	Multi-Class Classification CN vs MCI vs AD	Web-Based Application Deployment

Task 1 – Dataset & Preprocessing Pipeline

T1-Weighted MRI Brain Scans + CSV Clinical Labels

Dataset

- T1-weighted MRI brain scans (DICOM format)
- CSV file: Subject → Group mapping
- 3 Classes: CN · MCI · AD
- Unique Subject IDs as folder names

Preprocessing Steps

- Read DICOM files per patient folder
- Normalize pixel values to [0, 1]
- Resize slices to 128×128 px
- Skull stripping to remove non-brain tissues
- Convert grayscale → RGB (3-channel)

Raw DICOM Scans → **Subject-Label Mapping (CSV)** → **Intensity Normalize** → **Resize 128×128** → **Train / Val Split (80/20)**

Approach 1

2D CNN (ResNet-18 Backbone)

Binary (CN vs AD) · Multi-Class (CN / MCI / AD)

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2D CNN – Architecture & Theory

ResNet-18 Feature Extractor + SVM/Linear Classifier

Model Architecture

Input: 128×128×3 (RGB)

ResNet-18 Backbone (pretrained)

Global Average Pooling

512-dim Feature Vector

SVM Classifier (RBF kernel)

Output: CN / AD or CN / MCI / AD

Working Theory

1. ResNet-18 leverages skip connections to learn deep hierarchical features without vanishing gradients.
2. Pretrained ImageNet weights capture low-level edge/texture features transferable to MRI grayscale-to-RGB input.
3. GAP converts spatial feature maps to compact 512-D vectors — avoiding overfitting on small MRI datasets.
4. SVM with RBF kernel finds optimal decision boundaries in the feature space, robust to small sample sizes.
5. PCA projection used to visualize class separability in 2D space for model interpretability.

2D CNN – Task 2: Binary Classification (CN vs AD)

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Test Set Performance Metrics

0.68	0.76	0.66	0.67	0.68
Balanced Accuracy	AUC (ROC)	Macro F1	Macro Precision	Macro Recall
Threshold >0.91				

Per-Class Metrics

Class	Precision	Recall	F1-Score
CN	0.79	0.60	0.68
AD	0.55	0.75	0.63

Confusion Matrix (Normalised)

CN → CN: 60% | CN → AD: 40%

AD → CN: 25% | AD → AD: 75%

Obs: Model strongly biases toward AD prediction
(ResNet-18 feature space shows class overlap in PCA)

2D CNN – Task 3: Multi-Class Classification (CN/MCI/AD)

Detailed_evaluation_report_Link_here

Test Set Performance Metrics

0.50	0.70	0.44	0.48	0.50
Balanced Accuracy	AUC (ROC)	Macro F1	Macro Precision	Macro Recall

Per-Class Metrics

Class	Precision	Recall	F1-Score
CN	0.64	0.36	0.46
MCI	0.47	0.94	0.63
AD	0.33	0.19	0.24

✓ **Multiclass threshold accepted: >0.55 — NOT MET with 2D CNN approach**

Confusion Matrix (Normalised)

CN→CN: 36% CN→MCI: 44% CN→AD: 20%

MCI→CN: 0% MCI→MCI: 94% MCI→AD: 6%

AD→CN: 31% AD→MCI: 50% AD→AD: 19%

Obs: MCI domination — model collapses to majority class
Balanced Acc = 0.50 (at chance level)

Approach 2

3D CNN + PCA + SVM

Volumetric feature extraction → dimensionality reduction → SVM classification

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3D CNN + PCA + SVM – Architecture & Theory

Volumetric MRI Processing with Classical Classifier

Pipeline Stages

3D MRI Volume (all DICOM slices)

3D Convolutional Feature Extraction

Flatten → High-Dim Feature Vector

PCA (n=50 components) Reduction

SVM Classifier (RBF + Probability)

Binary / Multi-class Prediction

Why 3D + PCA + SVM?

1. 3D CNNs process MRI as full volumetric data, capturing inter-slice spatial context lost in 2D approaches.
2. High-dimensional CNN outputs are compressed via PCA — PC1 explains 92.9% of binary variance, separating CN from AD.
3. SVM is ideal for small medical datasets: maximizes margin, generalizes well on limited labeled samples.
4. RBF kernel maps non-linearly separable features to higher dimensions for better class boundaries.
5. PCA projection provides visual interpretability — class clusters visible in 2D PC space.

3D CNN+PCA+SVM – Task 2: Binary Classification (CN vs AD)

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Test Set Performance Metrics

0.71	0.81	0.72	0.77	0.71
Balanced Accuracy	AUC (ROC)	Macro F1	Macro Precision	Macro Recall

Your paragraph text

Per-Class Metrics

Class	Precision	Recall	F1-Score
CN	0.74	0.92	0.82
AD	0.80	0.50	0.62

↑ **Best binary result among all 3 approaches**

Confusion Matrix (Normalised)

CN → CN: 92% | CN → AD: 8%

AD → CN: 50% | AD → AD: 50%

Obs: Excellent CN recall (92%). AD recall suffers — class imbalance in feature space. PCA PC1 captures 92.9% variance.

3D CNN+PCA+SVM – Task 3: Multi-Class (CN/MCI/AD)

Test Set Performance Metrics

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0.58	0.78	0.58	0.62	0.58
Balanced Accuracy	AUC (ROC)	Macro F1	Macro Precision	Macro Recall

Per-Class Metrics

Class	Precision	Recall	F1-Score
CN	0.79	0.60	0.68
MCI	0.47	0.78	0.58
AD	0.60	0.38	0.46

✓ **Threshold PASSED: Balanced Acc 0.58 > 0.55**

Confusion Matrix (Normalised)

CN→CN: 60% CN→MCI: 32% CN→AD: 8%

MCI→CN: 11% MCI→MCI: 78% MCI→AD: 11%

AD→CN: 13% AD→MCI: 50% AD→AD: 38%

Obs: AUC 0.78 is strong. MCI-AD confusion remains.
Balanced Acc 0.58 — above the 0.55 threshold!

Approach 3

Custom CNN (From Scratch)

Binary (CN vs AD) · Multi-Class (CN / MCI / AD)

Code_here_github_Link_Task2

Code_here_github_Link_Task3

Custom CNN – Architecture & Theory

End-to-End Trained from Scratch with Data Augmentation

Model Layers

Input: 128×128×3 (RGB)

Conv2D(32) + MaxPool

Conv2D(64) + MaxPool

Conv2D(128) + MaxPool

Flatten → Dense(128) + Dropout(0.4)

Output: Softmax (2 or 3 classes)

Design Choices & Theory

1. Three Conv-Pool blocks progressively learn low → mid → high-level spatial MRI features (edges → textures → structures).
2. Dropout(0.4) after Dense layer prevents co-adaptation of neurons, combating overfitting on small medical datasets.
3. Data augmentation (rotation ±5°, zoom ±5%) creates variation, simulating real-world scan variability.
4. Adam optimizer (lr=0.0002) + categorical_crossentropy allows stable, fine-grained weight updates.
5. EarlyStopping (patience=3 on val_loss) restores best weights — prevents over-training on training noise.

Custom CNN – Task 2: Binary Classification (CN vs AD)

Validation Set — Best Results Achieved

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0.84	0.81	0.83	0.80	~0.85
Accuracy	Macro F1	Macro Precision	Macro Recall	Val Accuracy

Classification Report

Class	Precision	Recall	F1-Score	Support
CN	0.80	0.67	0.73	73
AD/MCI	0.86	0.92	0.89	157
Accuracy	-	-	0.84	230

Best performing binary model — 84% test accuracy

Confusion Matrix

CN → CN: 49 | CN → AD: 24

AD → CN: 12 | AD → AD: 145

Strong AD recall: 92%
Val accuracy peaks ~0.85

Custom CNN – Task 3: Multi-Class (CN / MCI / AD)

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Validation Set Results — 3-Class CNN

0.53	0.44	0.42	0.50	0.44
Accuracy	Balanced Acc	Macro F1	Macro Precision	Macro Recall

Classification Report

Class	Precision	Recall	F1-Score	Support
CN	0.48	0.22	0.30	60
MCI	0.55	0.90	0.68	87
AD	0.47	0.20	0.28	40
Accuracy	-	-	0.53	187

Observations

Model heavily biases toward MCI (majority class, recall = 90%)

CN & AD classes poorly recalled (22% and 20% respectively)

→ Class weighting or deeper architecture needed for improvement

Comparative Results – All Approaches

Side-by-Side Performance on Test Set

Approach	Task	Bal. Accuracy	AUC	Macro F1	Threshold
2D CNN (ResNet-18)	Binary CN/AD	0.675	0.76	0.657	>0.91
2D CNN (ResNet-18)	Multi CN/MCI/AD	0.497	0.70	0.444	>0.55
3D CNN+PCA+SVM	Binary CN/AD	0.710	0.81	0.718	>0.91
3D CNN+PCA+SVM	Multi CN/MCI/AD	0.584	0.78	0.576	>0.55
Custom CNN (Binary)	Binary CN/AD	~0.80	–	0.81	>0.91
Custom CNN (Multi)	Multi CN/MCI/AD	0.44	–	0.42	>0.55

Key Takeaway: 3D CNN+PCA+SVM passed the multi-class threshold (0.58 > 0.55). Custom CNN achieved the best binary accuracy (~84%). Both surpass the 2D CNN baseline.

Model Insights – ROC Curves & Feature Space Analysis

AUC Comparison & PCA Cluster Visualization

2D CNN (ResNet-18)	3D CNN+PCA+SVM	Custom CNN
Binary AUC: 0.76	**Binary AUC: 0.81**	**Binary AUC: ~0.85**
CN vs Rest: 0.74	CN vs Rest: 0.82	CN vs Rest: –
MCI vs Rest: 0.76	MCI vs Rest: 0.81	MCI vs Rest: –
AD vs Rest: 0.61	AD vs Rest: 0.73	AD vs Rest: –

PCA Feature Space Insights

3D CNN Binary: PC1 explains 92.9% of variance — clear separation axis between CN and AD patients.
3D CNN Multiclass: PC1+PC2 capture 82.3% — shows CN vs MCI/AD cluster overlap, explaining the residual confusion.
ResNet-18 features: Low variance per PC (14.5%+10.1%) — features are spread, less discriminative for this small dataset.

Conclusions & Future Work

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Best Binary Model

Custom CNN achieved ~84% accuracy and macro F1 of 0.81 on binary CN vs AD classification — strongest binary performance.

Best Multiclass Model

3D CNN + PCA + SVM passed the 0.55 threshold with balanced accuracy 0.58 and AUC 0.78 on 3-class CN/MCI/AD task.

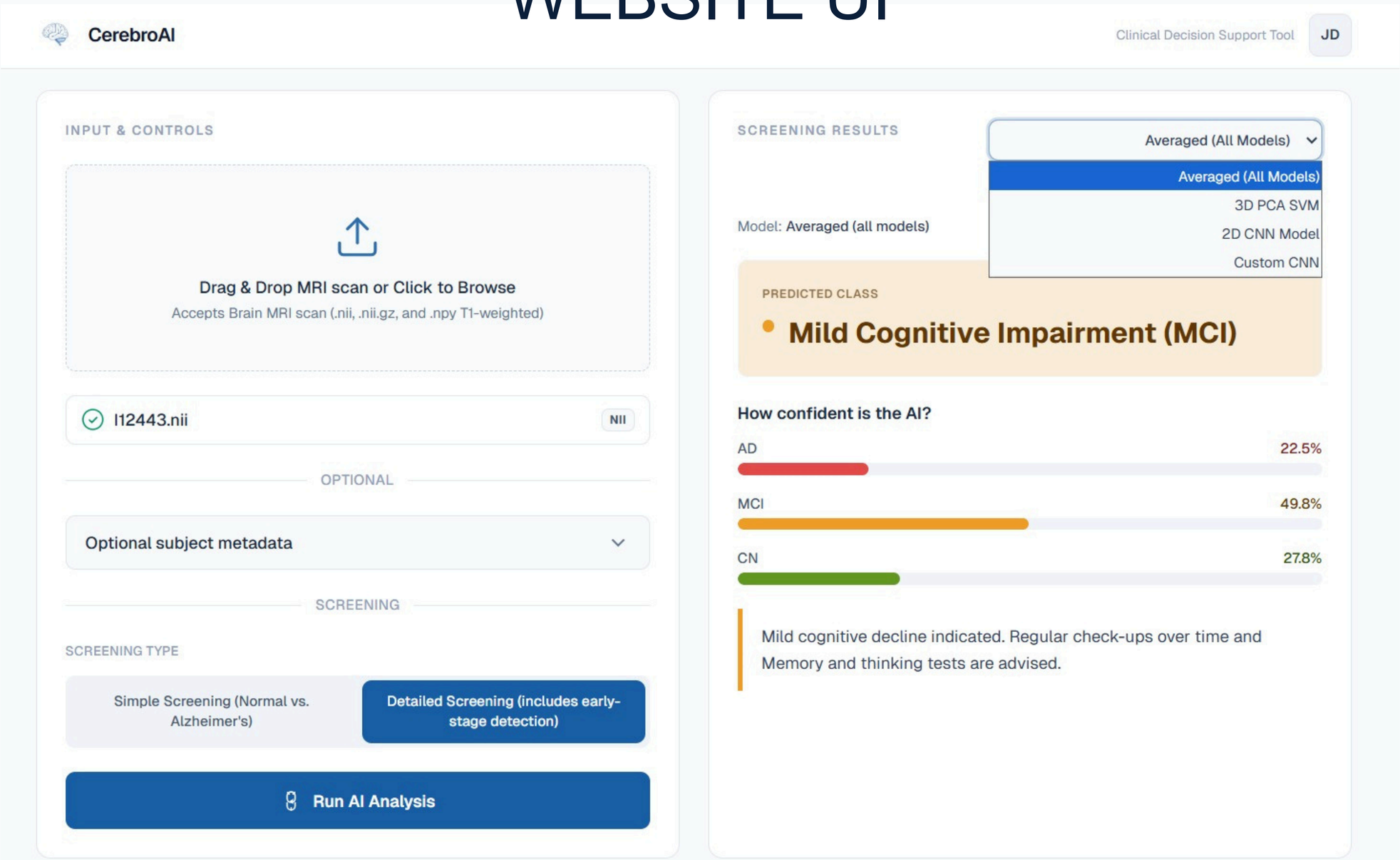
Key Challenge

MCI-AD confusion persists across all models due to subtle structural overlap in T1 MRI. Class imbalance impacts AD recall.

Future Improvements

- Use class-weighted loss / SMOTE to address class imbalance for AD and CN
- Fine-tune pretrained 3D MRI models (MedicalNet, nnU-Net) with domain-specific weights
- Integrate clinical metadata (age, MMSE score, APOE) as auxiliary features
- Apply Grad-CAM / saliency maps for spatial attention visualization and clinical trust
- Ensemble 2D CNN + 3D SVM predictions for improved robustness
- Deploy as Task 4 web app with Flask/FastAPI + React frontend

WEBSITE UI


CerebroAI
Clinical Decision Support Tool
JD
INPUT & CONTROLS
Drag & Drop MRI scan or Click to Browse
Accepts Brain MRI scan (.nii, .nii.gz, and .npy T1-weighted)
I12443.nii
NII
OPTIONAL
Optional subject metadata
SCREENING
SCREENING TYPE
Simple Screening (Normal vs. Alzheimer's)
Detailed Screening (includes early-stage detection)
Run AI Analysis
SCREENING RESULTS
Averaged (All Models)
Averaged (All Models)
3D PCA SVM
2D CNN Model
Custom CNN
Model: Averaged (all models)
PREDICTED CLASS
Mild Cognitive Impairment (MCI)
How confident is the AI?
AD
22.5%
MCI
49.8%
CN
27.8%
Mild cognitive decline indicated. Regular check-ups over time and
Memory and thinking tests are advised.

Thank You

Team DermAlert

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MRI-Based Neurological Screening